Recurrence Relation:

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

$a \geq 1 \quad b \geq 2$

$f(n)$ is any mathematical function

```
int func(int n) {
    if (n==1) {          ———————— 1
        return;          ———————— 1
    }
    func(n/2);           ———————— T(n/2)
}
```

Now,

$T(n) = T\left(\frac{n}{2}\right) + 1$ ——— M

put $n = n/2$ in M

$T(n/2) = T(n/2^2) + 1$

put $T(n/2)$ in M

$T(n) = T(n/2^2) + 2$ ——— (i)

Put $n = n/2^2$ in M

$T(n/2^2) = T(n/2^3) + 1$

Put in (i)

$T(n) = T(n/2^3) + 3$

⋮ k^{th} iteration

$T(n) = T(n/2^k) + k$

This goes on till $\frac{n}{2^k} = 1$

$k = \log_2(n)$

$T(n) = T\left(\frac{n}{2^{\log_2 n}}\right) + \log_2 n$

$= T(1) + \log_2 n$

$= 1 + \log_2(n)$

$O(\log_2 n)$

Matrix Addition

We divide the matrix into $n/2 \times n/2$ blocks till we have a single element, then we add them

$$A = \left[\begin{array}{cc|cc} 1 & 2 & 3 & 4 \\ & a_{11} & & a_{12} \\ 5 & 6 & 7 & 8 \\ \hline 9 & 10 & 11 & 12 \\ & a_{21} & & a_{22} \\ 13 & 14 & 15 & 16 \end{array}\right]_{4\times 4}$$

$$B = \left[\begin{array}{cc|cc} 1 & 2 & 3 & 4 \\ & b_{11} & & b_{12} \\ 5 & 6 & 7 & 8 \\ \hline 9 & 10 & 11 & 12 \\ & b_{21} & & b_{22} \\ 13 & 14 & 15 & 16 \end{array}\right]_{4\times 4}$$

$$C = \left[\begin{array}{c|c} \begin{array}{c|c} c^*_{11} = a^*_{11} + b^*_{11} & c^*_{12} = a^*_{12} + b^*_{12} \\ \hline c^*_{21} = a^*_{21} + b^*_{21} & c^*_{22} = a^*_{22} + b^*_{22} \end{array} \; C_{11} = a_{11} + b_{11} & C_{12} = a_{12} + b_{12} \\ \hline C_{21} = a_{21} + b_{21} & C_{22} = a_{22} + b_{22} \end{array}\right]_{4\times 4}$$

```
int Matrix Addition (A, B, C, n) {
    if (n == 1) {
        return a* + b*;
    }

    C11 = Matrix Addition (A, B, C, n/2);
    C12 = Matrix Addition (A, B, C, n/2);
    C21 = Matrix Addition (A, B, C, n/2);
    C22 = Matrix Addition (A, B, C, n/2);
}
```

$T(n) = 4T(n/2) + 1$ —— M

put $n = n/2$ in M

$T(n/2) = 4T(n/2^2) + 1$

put $T(n/2)$ in M

$T(n) = 4^2 T(n/2^2) + 4 + 1$

put $n = n/2^2$ in M

$T(n/2^2) = 4T(n/2^3) + 1$

put $T(n/2^2)$ in (i)

$T(n) = 4^3 T(n/2^3) + 4^2 + 4 + 1$

⋮ kth iteration

$$T(n) = 4^k T(n/2^k) + 4^{k-1} + \dots + 4^2 + 4 + 1$$

Base case reached when

$$\frac{n}{2^k} = 1$$

$$k = \log_2(n)$$

$$T(n) = 4^{\log_2 n} T\left(\frac{n}{2^k}\right) + \frac{4^k - 1}{3}$$

$$T(n) = 4^{\log_2 n} T\left(\frac{n}{2^{\log n}}\right) + \frac{4^{\log_2 n} - 1}{3}$$

$$= 2^{2\log_2 n} T\left(\frac{n}{2^{\log n}}\right) + \frac{2^{2\log n} - 1}{3}$$

$$= \cancel{2}^{\cancel{\log_2} n^2} T\left(\frac{\cancel{n}}{\cancel{2^{\log n}}}\right) + \frac{\cancel{2}^{\cancel{\log} n^2} - 1}{3}$$

$$= n^2 T(1) + \frac{n^2 - 1}{3}$$

$$= n^2 + \frac{n^2 - 1}{3}$$

$O(n^2)$

Day: | Date: / /

Example,

$$T(n) = 2T\left(\frac{n}{2}\right) + n \text{ --- } ①$$

put $n = \frac{n}{2}$ in ①

$$T(n/2) = 2T\left(\frac{n}{2^2}\right) + \frac{n}{2}$$

$$T(n) = 2^2 T\left(\frac{n}{2^2}\right) + n + n$$

Put $n = \frac{n}{2^2}$ in ①

$$T\left(\frac{n}{2^2}\right) = 2T\left(\frac{n}{2^3}\right) + \frac{n}{2^2}$$

$$T(n) = 2^3 T\left(\frac{n}{2^3}\right) + 3n$$

k^{th} iter

$$T(n) = 2^k T\left(\frac{n}{2^k}\right) + kn$$

Terminate when $\frac{n}{2^k} = 1$

$k = \log n$

$$T(n) = 2^{\log_2 n} T\left(\frac{n}{2^{\log_2 n}}\right) + \log_2 n \; n$$

$$= nT(1) + n\log_2 n$$

$$= n + n\log_2 n \sim O(n\log_2 n)$$

GALAXAY

$T(n) = T(n/2) + n$ —— M

put $n = n/2$ in M

$T(n/2) = T(n/2^2) + \frac{n}{2}$

put $T(n/2)$ in M

$T(n) = T\left(\frac{n}{2^2}\right) + \frac{n}{2} + n$ —— (i)

put $n = n^2/2$ in M

$T\left(\frac{n}{2^2}\right) = T(n/2^3) + \frac{n}{2^2}$

put $T(n/2^2)$ in (i)

$T(n) = T\left(\frac{n}{2^3}\right) + \frac{n}{2^2} + \frac{n}{2} + n$

$\vdots$ kth

$T(n) = T\left(\frac{n}{2^k}\right) + \frac{n}{2^{k-1}} + \dots + n$

$\frac{n}{2^k} = 1$

$k = \log n$

$T(n) = T\left(\frac{n}{2^k}\right) + n\left(\frac{1}{2^{k-1}} + \dots + \frac{1}{2} + 1\right)$

$= T\left(\frac{n}{2^k}\right) + 2n$

$= T\left(\frac{n}{2^{\log_2 n}}\right) + 2n$

$= T(1) + 2n$

$O(n)$